

Comparison with Viglione et al. (2013)

Viglione et al. (2013) and (Skahill, Viglione, & Byrd, 2016) present a Bayesian analysis framework for combining and evaluating the worth of different types of additional data (i.e., temporal, spatial, and causal) in a flood frequency analysis. Temporal information expansion is directed toward collecting information on the flood behavior before (or after) the systematic data period (Viglione, Merz, Salinas, & Blöschl, 2013). Spatial information expansion is based on using flood information from neighboring catchments to improve flood frequency estimates at the site of interest. Causal information expansion analyzes the generating mechanisms of floods in the catchment of interest (Merz & Blöschl, 2008).

Temporal information is added to the Bayesian analysis through the inclusion of interval- and threshold-censored data in the likelihood function as discussed in the Maximum Likelihood Estimation (MLE) section. Regional information on distribution parameters can be incorporated through the use of informative priors, as shown in the Informative Priors section. Causal information is incorporated into the analysis by first defining the prior distribution for a flood quantile:

$$h(Q_p) = N(\mu_p, \sigma_p) \quad \text{Equation 23}$$

where Q_p is the flood discharge for a specific annual exceedance probability P ; and $h(\cdot)$ is the PDF of the Normally distributed variable Q_p . Q_p is determined from the inverse cdf $F^{-1}(\cdot)$ of the parent distribution conditional on the parameters:

$$Q_p = F^{-1}(P|\theta) \quad \text{Equation 24}$$

The inverse CDF is then plugged into the PDF of the quantile prior to get:

$$\pi(\theta) = h(F^{-1}(P|\theta)) \quad \text{Equation 25}$$

During the Bayesian MCMC routine, the likelihood of the various components are multiplied by the quantile prior $\pi(\theta)$ to calculate a posterior distribution of the parameters, which is consistent with a reasonable range of Q_p . The overall likelihood function is then constructed by multiplying all of the components:

$$L(D|\theta) = L_S(D|\theta) \cdot L_L(D|\theta) \cdot L_I(D|\theta) \cdot \pi(\theta) \quad \text{Equation 26}$$

It can be seen in Equation 26 that the causal indicator $\pi(\theta)$ behaves similarly to a penalty function, in that it rewards parameter sets that produce a reasonable likelihood of Q_p and discounts those that do not. All other things being equal, parameter sets that do not agree well with $h(Q_p)$ will have lower likelihoods than those that do.

Skahill et al. (2016) independently revisited components of the example originally profiled by (Viglione, Merz, Salinas, & Blöschl, 2013), performing eight distinct MCMC simulations using the Kamp at Zwettl dataset. A summary of the simulations performed are provided in Table 80.

Table 80 – Summary of the Eight Distinct MCMC Simulation (Skahill, Viglione, & Byrd, 2016).

MCMC Simulation	Data of the Kamp at Zwettl
1	Systematic data (1951-2001)
2	Systematic data (1951-2005)
3	Systematic data (1951-2001) + temporal information expansion
4	Systematic data (1951-2005) + temporal information expansion
5	Systematic data (1951-2001) + causal information expansion
6	Systematic data (1951-2005) + causal information expansion
7	Systematic data (1951-2001) + temporal + causal information expansion
8	Systematic data (1951-2005) + temporal + causal information expansion

For each MCMC simulation, the posterior mode (PM) estimate for the GEV parameters and the 100-yr and 1,000-yr discharges, including the 90% credible interval, were provided in (Skahill, Viglione, & Byrd, 2016). The same eight simulations were performed using RMC-BestFit and compared with the results from Skahill et al. (2016). The default flat priors for the GEV were used with 100,000 posterior parameter sets. Results are provided in Table 81 through Table 83. RMC-BestFit produces nearly identical results for each scenario.

Table 81 – Comparison RMC-BestFit with (Skahill, Viglione, & Byrd, 2016) for Posterior Mode Parameters for the GEV Distribution.

MCMC Simulation	Skahill et al. (2016) - GEV Parameters			RMC-BestFit - GEV Parameters		
	Location (ξ)	Scale (α)	Shape (κ)	Location (ξ)	Scale (α)	Shape (κ)
1	42.9	20.2	-0.096	42.9	20.2	-0.096
2	41.7	20.7	-0.310	41.8	20.7	-0.310
3	43.4	21.7	-0.222	43.3	21.6	-0.221
4	42.6	21.5	-0.281	42.7	21.5	-0.279
5	41.9	21.0	-0.313	41.8	21.0	-0.312
6	41.6	20.8	-0.333	41.6	20.8	-0.333
7	42.7	21.8	-0.291	42.7	21.8	-0.292
8	42.5	21.5	-0.313	42.5	21.6	-0.311

Table 82 – Comparison RMC-BestFit with (Skahill, Viglione, & Byrd, 2016) for the 100-yr Quantile.

MCMC Simulation	Skahill et al. (2016) - Q_{100} (m ³ /s)			RMC-BestFit - Q_{100} (m ³ /s)		
	PM	5%	95%	PM	5%	95%
1	160	130	288	160	128	288
2	253	184	542	253	184	545
3	217	176	291	216	176	291
4	244	197	331	244	197	331
5	258	193	307	257	192	307
6	269	217	317	269	216	317
7	253	206	299	254	207	299
8	264	220	308	263	220	308

Table 83 – Comparison RMC-BestFit with (Skahill, Viglione, & Byrd, 2016) for the 1,000-yr Quantile.

MCMC Simulation	Skahill et al. (2016) - Q_{1000} (m ³ /s)			RMC-BestFit - Q_{1000} (m ³ /s)		
	PM	5%	95%	PM	5%	95%
1	241	183	649	240	163	654
2	543	317	1853	544	310	1952
3	399	278	647	396	274	656
4	497	347	818	495	339	836
5	557	335	702	554	329	725
6	604	418	747	604	411	768
7	527	369	671	529	362	684
8	571	418	708	568	410	725

The method for incorporating a quantile prior proposed by Viglione et al. (2013), and described above, is similar to the weighting of two independent and Normally distributed estimates using Equation 21 and Equation 22 from the Informative Priors section. Accordingly, a comparison was performed between RMC-BestFit and the theoretical weighting method.

Simulation #1 used the systematic Kamp at Zwettl dataset for years 1951-2001. Simulation #3 combined the systematic dataset with historical (temporal) information dating back to year 1600. The mean and standard deviation of the at-site 500-year quantile for the two datasets are shown in Table 84.

In Viglione et al. (2013), expert elicitation was used to determine a prior for the 500-year quantile (0.002 exceedance probability), which was set as a Normal distribution with mean of 480 m³/s and standard deviation of 80 m³/s. Simulation #5 was performed using the systematic data plus the quantile prior. Simulation #7 used the systematic data plus the historical data and the quantile prior. Results for each dataset are provided in Table 85 and Table 86 and illustrated in Figure 49 and Figure 50, respectfully.

The results from RMC-BestFit agree reasonably well with the theoretical approximation. However, recall that the theoretical approximation assumes the two estimates are Normally distributed. It can be seen in Figure 49 that the quantile distribution for simulation #1 (systematic data only) exhibits considerable skewness. Therefore, it would be unreasonable to expect the theoretical weighting solution to match RMC-BestFit since the Normality assumption is not satisfied. As the at-site data increases, the Normality assumption becomes more reasonable and the theoretical approximation can be expected to perform better. Figure 50 shows the quantile distribution for simulation #3 (historical data), which has less skewness than simulation #1. As expected, the theoretical solution more closely matches RMC-BestFit for this dataset as shown in Table 86.

Table 84 – Summary Statistics for the At-Site 500-Year Quantile for Kamp at Zwettl.

Q_{500} (m ³ /s)	Simulation #1	Simulation #3
Mean (μ)	264.67	350.73
Std. Deviation (σ)	148.02	87.93

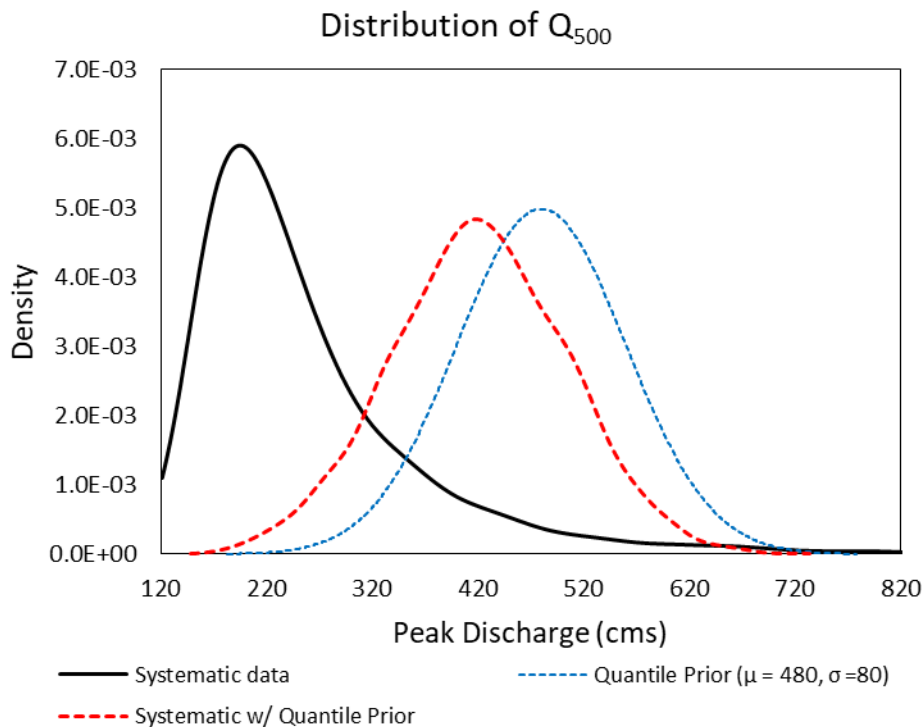


Figure 49 – Distributions of Quantile Q_{500} in RMC-BestFit for MCMC Simulation #5.

Table 85 – Posterior Quantile Statistics in RMC-BestFit Compared to the Theoretical Solution for MCMC Simulation #5.

Q_{500} (m ³ /s)	Theoretical	RMC-BestFit
Mean (μ)	431.32	421.37
Std. Deviation (σ)	70.38	83.75

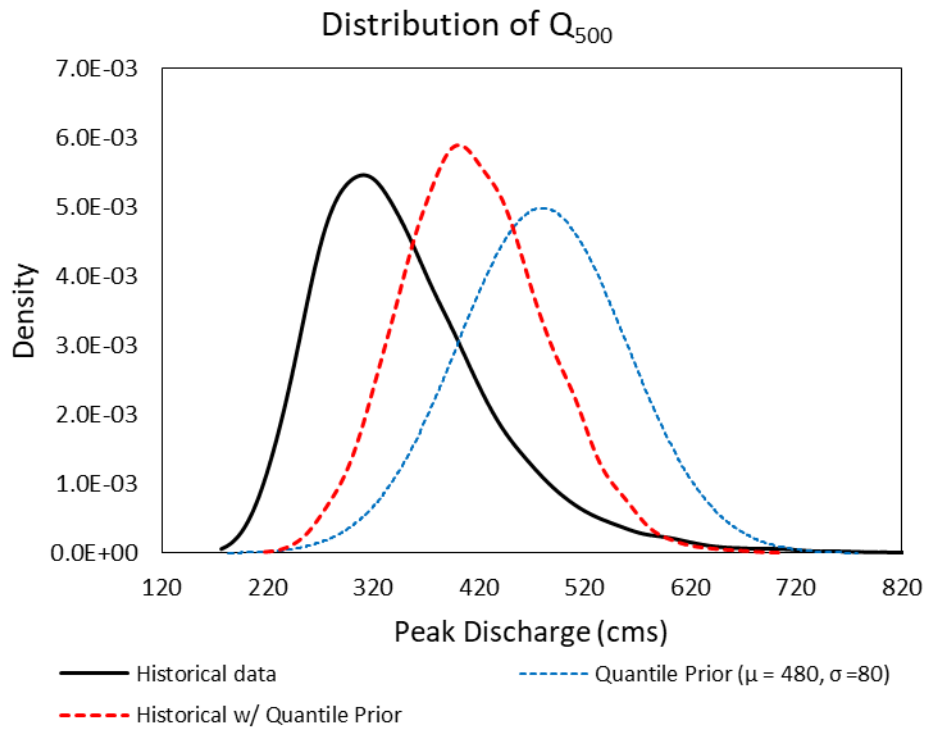

 Figure 50 – Distributions of Quantile Q_{500} in RMC-BestFit for MCMC Simulation #7.

Table 86 – Posterior Quantile Statistics in RMC-BestFit Compared to the Theoretical Solution for MCMC Simulation #7.

Q_{500} (m ³ /s)	Theoretical	RMC-BestFit
Mean (μ)	421.46	416.10
Std. Deviation (σ)	59.17	67.65

Comparison with Evdbayes

The *evdbayes* package is an add-on package for the R programming environment that provides functions for the Bayesian analysis of extreme value models using MCMC (Stephenson & Ribatet, 2006). The *evdbayes* package provides several options for priors, including an option for setting priors on distribution quantiles following the approach used in (Coles & Tawn, 1996). This same approach is also implemented in RMC-BestFit.

A prior distribution can be constructed in terms of the quantiles (q_{p1}, q_{p2}, q_{p3}) for specified exceedance probabilities $p_1 < p_2 < p_3$. Since $q_{p1} > q_{p2} > q_{p3}$ it is easier to work with differences $(\bar{q}_{p1}, \bar{q}_{p2}, \bar{q}_{p3})$, where $\bar{q}_{pi} = q_{pi} - q_{pi-1}$. The number of exceedance probabilities must be equal to the number of distribution parameters. For example, for the GEV distribution there must be three quantile priors with three quantile differences. The priors on the quantile differences are assumed to be independent and Gamma distributed with $\bar{q}_{p1} \sim \text{Gamma}(\alpha_i, \beta_i)$. This leads to the prior density for the GEV quantiles as:

$$\pi(\theta) = |J| \cdot \prod_{i=1}^3 \bar{q}_{pi} \quad \text{Equation 27}$$

where J is the Jacobian of the transformation from (q_{p1}, q_{p2}, q_{p3}) to $\theta = (\xi, \alpha, \kappa)$. More details on this method can be found in (Coles & Tawn, 1996) and (Smith, 2005). This approach for quantile priors has been generalized to work for any of the thirteen probability distributions offered in RMC-BestFit.

First, before verifying the method for quantile priors, a comparison was performed with uninformative priors using the Kamp at Zwetl dataset from 1951-2001 (Viglione, Merz, Salinas, & Bloschl, 2013). Example code for *evdbayes* is provided in Figure 51 so that others will be able to reproduce these results. The *evdbayes* package provides summary statistics for the GEV parameters, return period plots, and the posterior predictive distribution. For this comparison, the default flat priors were used in RMC-BestFit with 100,000 posterior parameter sets. In *evdbayes*, the priors for parameters are set to be a multivariate Normal distribution with very large variances in order to make them uninformative. Summary statistics of the verification results are provided in Table 87 and Table 88, and a frequency curve plot comparing curves is provided in Figure 52. It can be seen that RMC-BestFit and *evdbayes* produce effectively identical results. In addition, Figure 52 provides further confirmation that RMC-BestFit is correctly computing the posterior predictive distribution.

Table 87 – Parameter Summary Statistics from Evdbayes with Uninformative Priors.

Parameter	Mean	Std. Dev.	2.5%	50%	97.5%
Location (ξ)	42.3582	3.335	35.98100	42.2910	49.1315
Scale (α)	20.9698	2.703	16.31164	20.7482	26.8899
Shape (κ)	-0.1252	0.129	-0.4056	-0.1154	0.9938

Table 88 – Parameter Summary Statistics from RMC-BestFit with Uninformative Priors.

Parameter	Mean	Std. Dev.	2.5%	50%	97.5%
Location (ξ)	43.0466	3.4184	36.6124	42.9615	49.9683
Scale (α)	21.5134	2.8006	16.6868	21.2830	27.6154
Shape (κ)	-0.1162	0.1312	-0.4020	-0.1065	0.1107

```

library(evd)
library(evdbayes)
library(coda)

# This is example code with evdbayes with the GEV distribution and uninformative priors.
# Kamp at Zwettl, 1951-2001. Dataset from Viglione (2013).
gageData = c(135, 52, 45, 95, 54, 94, 95, 56, 140, 72, 50.9, 54, 105, 53, 62, 51, 58, 39, 62, 68,
             44, 52, 36, 60, 100, 41.2, 60.2, 19.7, 50.6, 35.6, 46.1, 42, 26, 32.1, 89.1, 29,
             58.3, 46.4, 20.9, 17, 74.4, 22.6, 73.2, 41.8, 34.6, 120, 43, 23.7, 56.9, 30.4, 21.8)

# Estimate the posterior mode parameter set using Maximum Likelihood Estimation (MLE).
mle <- fgev(x=gageData, method="Nelder-Mead", std.err = FALSE)
print(mle)
# Estimates
#      loc      scale      shape
# 42.91251 20.20805 0.09625

# Set up MCMC inputs
gevCovMatrix <- diag(c(1000, 1000, 10))
gevPriors <- prior.norm(mean=c(0,0,0), cov=gevCovMatrix)
initialVals <- c(as.numeric(mle$param[1]), as.numeric(mle$param[2]), as.numeric(mle$param[3]))
gevPosteriors <- posterior(n=2000000, init=initialVals, prior=gevPriors, lh="gev",
                          data=gageData, psd=c(5,.1,.1), burn=60000, thin=as.integer(20))

# Run MCMC
gevMCMC <- mcmc(gevPosteriors)

# Output the summary statistics for the Bayesian estimated parameters.
summary(gevMCMC)
# Note: The sign of the shape parameter (xi) is reverse of the
# Hosking parameterization (kappa) used in RMC-BestFit.
#
#      Mean      SD Naive SE Time-series SE
# mu    42.3582  3.335 0.0107066    0.0113403
# sigma 20.9698  2.703 0.0086788    0.0095153
# xi     0.1252  0.129 0.0004143    0.0004477
#
#      2.5%      25%      50%      75% 97.5%
# mu    35.98100 40.09090 42.2910 44.5371 49.1315
# sigma 16.31164 19.06830 20.7482 22.6337 26.8899
# xi    -0.09938 0.03437 0.1154 0.2049 0.4056

# Plot Markov chain trace plots and kernel density estimates for each parameter.
plot(gevMCMC)

# Plot frequency curve with confidence intervals.
rl.pst(gevMCMC, lh="gev", ylim=c(10,2000), ci=0.95)
print(rl.pst(gevMCMC, lh="gev", ylim=c(10,2000), ci=0.95))

# Plot posterior predictive curve.
rl.pred(gevMCMC, lh="gev", qlim=c(10,3000))
print(rl.pred(gevMCMC, lh="gev", qlim=c(10,3000)))

```

Figure 51 – Example code for evdbayes with uninformative priors.

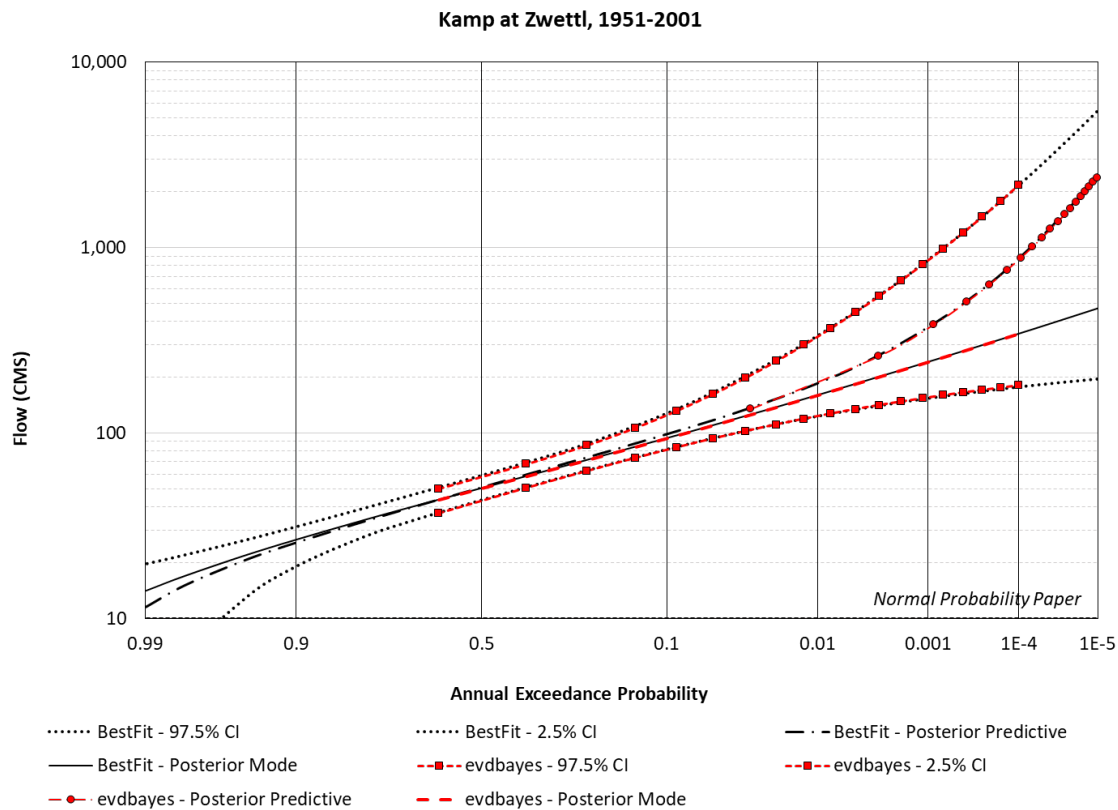


Figure 52 – Comparison of RMC-BestFit with Evdbayes with Uninformative Priors.

The next verification compared RMC-BestFit with *evdbayes* with informative priors on quantiles. In RMC-BestFit, the user specifies priors on quantiles using a Normal distribution because it is more intuitive than specifying the Gamma distributed quantile differences. The Gamma distributions for quantile differences are estimated automatically in the code behind. Using the mean and variance of the difference between two independent Normal distributions, the scale and shape parameters of the Gamma distribution are then estimated using the direct method of moments. Table 89 provides a summary of the prior distributions used for this example.

Table 89 – Summary of Prior Distributions on Quantiles.

Exceedance Probability	Normal Distribution for Quantiles		Gamma Distribution for Differences	
	Mean	Std. Dev.	Scale	Shape
0.1	100	20	4.00	25.00
0.01	250	40	13.33	11.25
0.001	500	60	20.80	12.02

Example code for *evdbayes* with informative priors is provided in Figure 54. Summary statistics of the verification results are provided in Table 90 and Table 91, and a frequency curve plot comparing curves is provided in Figure 53. RMC-BestFit and *evdbayes* produce nearly identical results.

Table 90 – Parameter Summary Statistics from Evdbayes with Informative Priors on Quantiles.

Parameter	Mean	Std. Dev.	2.5%	50%	97.5%
Location (ξ)	41.4307	3.0918	35.5680	41.3467	47.7391
Scale (α)	20.7700	2.4520	16.3861	20.6195	26.0375
Shape (κ)	-0.2694	0.0488	-0.3608	-0.2709	-0.1699

Table 91 – Parameter Summary Statistics from RMC-BestFit with Informative Priors on Quantiles.

Parameter	Mean	Std. Dev.	2.5%	50%	97.5%
Location (ξ)	41.4281	3.1001	35.5634	41.3612	47.7805
Scale (α)	20.7617	2.4513	16.4351	20.5992	25.9730
Shape (κ)	-0.2690	0.0490	-0.3614	-0.2703	-0.1689

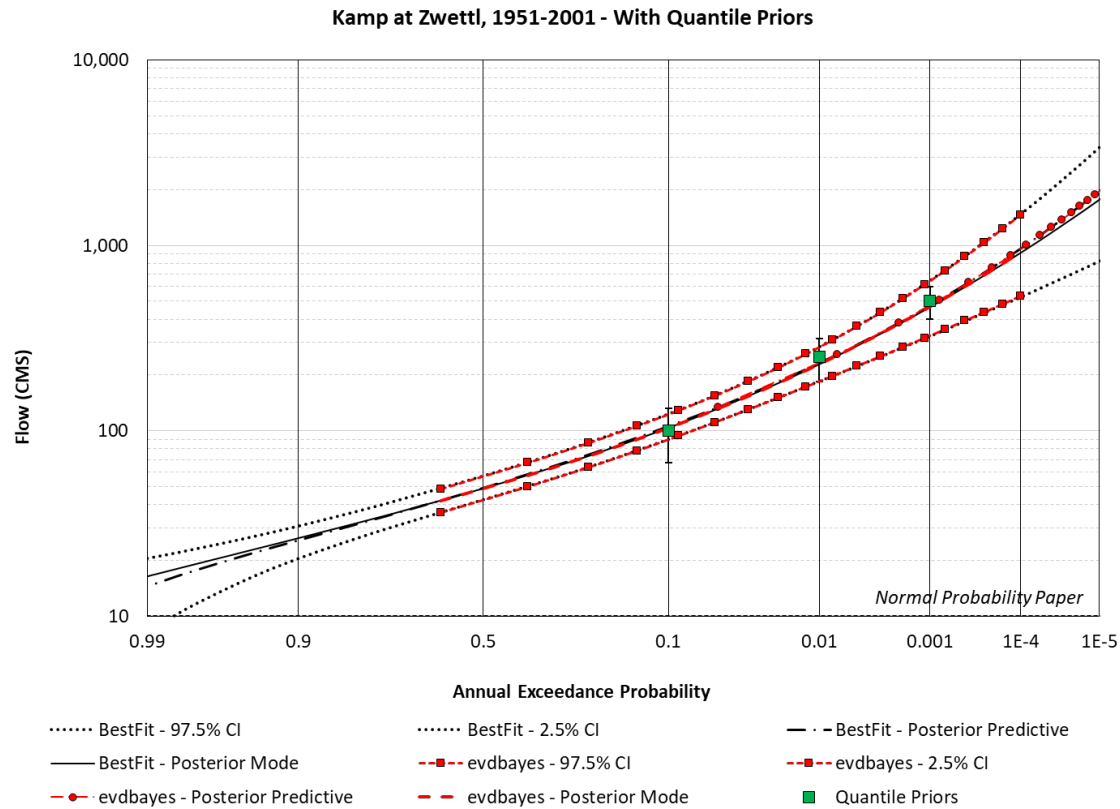


Figure 53 – Comparison of RMC-BestFit with Evdbayes with Informative Priors on Quantiles.


```

library(evd)
library(evdbayes)
library(coda)

# This is example code with evdbayes with the GEV distribution and Informative priors.
# Kamp at Zwettl, 1951-2001. Dataset from Viglione (2013).
gageData = c(135, 52, 45, 95, 54, 94, 95, 56, 140, 72, 50.9, 54, 105, 53, 62, 51, 58, 39, 62, 68,
             44, 52, 36, 60, 100, 41.2, 60.2, 19.7, 50.6, 35.6, 46.1, 42, 26, 32.1, 89.1, 29,
             58.3, 46.4, 20.9, 17, 74.4, 22.6, 73.2, 41.8, 34.6, 120, 43, 23.7, 56.9, 30.4, 21.8)

# Estimate the posterior mode parameter set using Maximum Likelihood Estimation (MLE).
mle <- fgev(x=gageData, method="Nelder-Mead", std.err = FALSE)
print(mle)
#      loc      scale  shape
# 42.91251 20.20805 0.09625

# Set up MCMC inputs
gevPriors <- prior.quant(shape=c(25,6.25,6.25), scale=c(4,24,40))
initialVals <- c(as.numeric(mle$param[1]), as.numeric(mle$param[2]), as.numeric(mle$param[3]))
gevPosteriors <- posterior(n=2000000, init=initialVals, prior=gevPriors, lh="gev",
                          data=gageData, psd=c(5,.1,.1), burn=60000, thin=as.integer(20))

# Estimate maximum a posteriori (MAP)
map <- mposterior(data=gageData, init=initialVals, prior=gevPriors, lh="gev", method="Nelder-Mead",
                  control=list(maxit=20000))
print(map)
# 41.2176265 19.9717628 0.2817834

# Run MCMC
gevMCMC <- mcmc(gevPosteriors)

# Output the summary statistics for the Bayesian estimated parameters.
summary(gevMCMC)
# Note: The sign of the shape parameter kappa is reverse of the
# Hosking parameterization used in RMC-BestFit.
#
#      Mean      SD Naive SE Time-series SE
# mu  41.4307 3.0918 0.0099271    0.0105737
# sigma 20.7700 2.4520 0.0078727    0.0086493
# xi    0.2694 0.0488 0.0001567    0.0001637
#
#      2.5%    25%    50%    75%    97.5%
# mu  35.5680 39.3004 41.3467 43.4675 47.7391
# sigma 16.3861 19.0696 20.6195 22.3158 26.0375
# xi    0.1699 0.2376 0.2709 0.3028 0.3608

# Plot Markov chain trace plots and kernel density estimates for each parameter.
plot(gevMCMC)

# Plot frequency curve with confidence intervals.
rl.pst(gevMCMC, lh="gev", ylim=c(10,2000), ci=0.95)
print(rl.pst(gevMCMC, lh="gev", ylim=c(10,2000), ci=0.95))

# Plot posterior predictive curve.
rl.pred(gevMCMC, lh="gev", qlim=c(10,3000))
print(rl.pred(gevMCMC, lh="gev", qlim=c(10,3000)))

```

Figure 54 – Example code for evdbayes with Informative Priors on Quantiles.